Stellar Object Classification

CMPS 470/570

TEAM: Group B
Members: Adam Rodi, Ron Logarbo
Github Link: https://github.com/adamrodi/CMPS4700_Group_B

TEAM MEMBERS, ROLES, and Participation

Adam Rodi

Contributions:

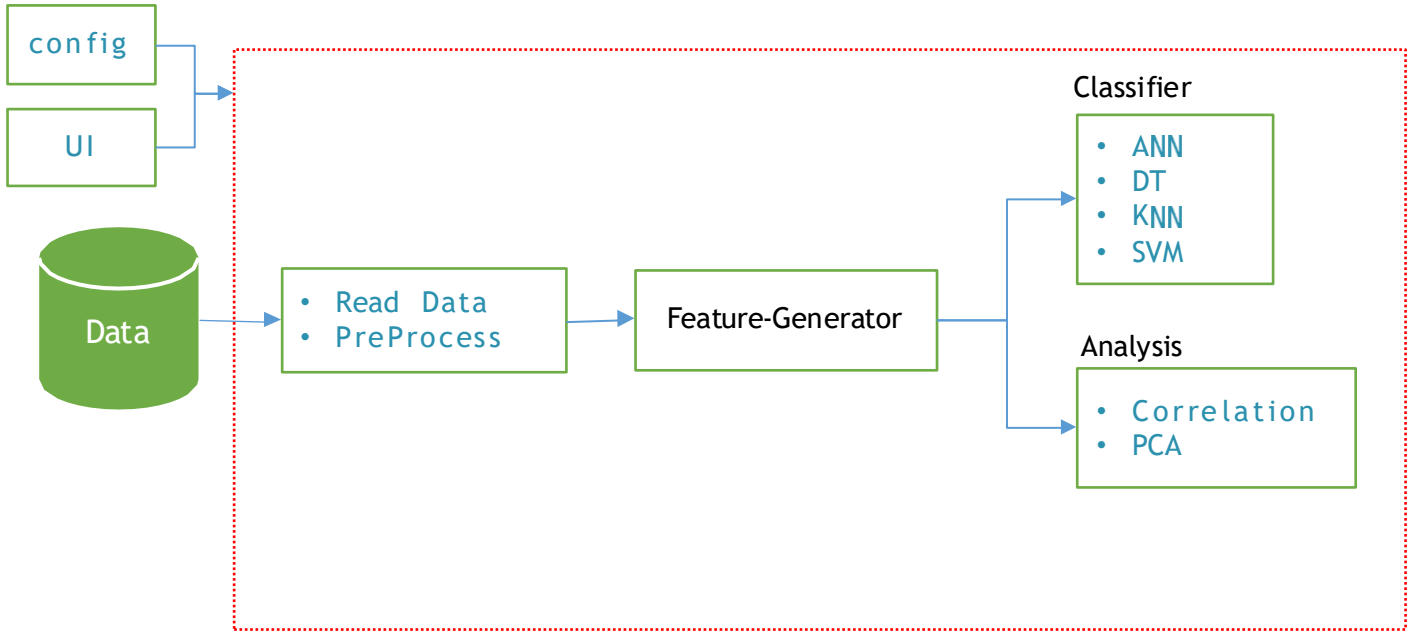
- Preprocessing: Stratified split design, z-score scaling pipeline, schema validation, identifier column removal.
- Feature Extraction: Pearson correlation matrix analysis, PCA component selection (95% variance threshold).
- Modeling: ANN and K-NN classifiers, model evaluation framework (metrics, confusion matrices).

Ron Logarbo

Contributions:

- Preprocessing: Excel/CSV export functions, preprocessing plots (class distribution, before/after scaling).
- Feature Extraction: PCA parameter export (JSON), scatter pair grid, features_data Excel output.
- Modeling: SVM and Decision Tree classifiers, performance comparison plots, model serialization.

MODULE COMMUNICATION GRAPH



- Export
  - Features
  - Model
  - Performance-epoch curve
  - Predictions
  - Performance scores
  - Plots
    - Performance-epoch curve
    - Performance scores

DESCRIPTION OF THE PROJECT

Goal

- Automatically classify astronomical objects as GALAXY, STAR, or QSO (quasar) using four ML algorithms: ANN, SVM, Decision Tree, and K-NN

Dataset

- Sloan Digital Sky Survey (SDSS DR17) — 100,000 labeled observations
- 8 photometric and positional features used after dropping 9 identifier columns

Pipeline Stages

- **Stage 1 — Preprocessing:** clean, split (60/20/20), z-score scale
- **Stage 2 — Feature Extraction:** correlation analysis, PCA (8 → 5 components, 99.6% variance)
- **Stage 3 — Classification:** train & evaluate ANN, SVM, DT, K-NN; ANN achieves 97.0% test accuracy

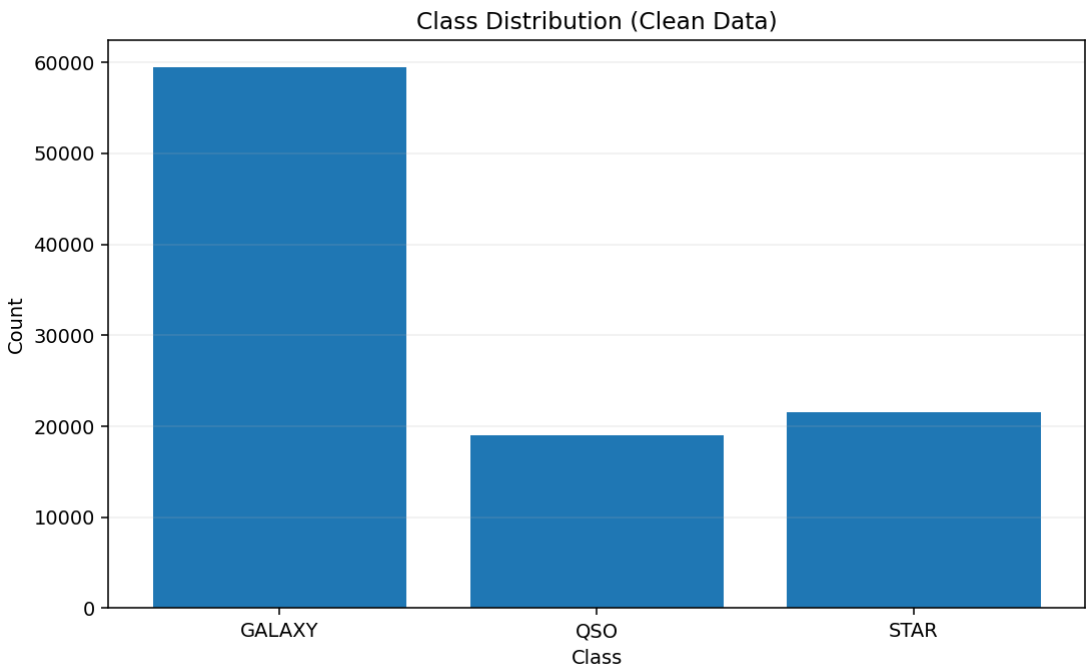
Tools

- Python, scikit-learn, pandas, numpy, matplotlib

DESCRIPTION OF THE RAW DATA

Feature	Type	Description
u	Float	Ultraviolet filter brightness (magnitude)
g	Float	Green filter brightness (magnitude)
r	Float	Red filter brightness (magnitude)
i	Float	Near-IR filter brightness (magnitude)
z	Float	Infrared filter brightness (magnitude)
alpha	Float	Right ascension — sky longitude (degrees)
delta	Float	Declination — sky latitude (degrees)
redshift	Float	Recession velocity proxy (0 ≈ star, >0 ≈ galaxy/QSO)
class	String	Target label: GALAXY, STAR, or QSO

Source: SDSS Data Release 17 (DR17) — 100,000 observations



Class distribution: GALAXY 59,445 (59.4%) | STAR 21,594 (21.6%) | QSO 18,961 (19.0%)

Original columns: 18 (9 dropped identifiers, 8 features retained + 1 class label)

DESCRIPTION OF THE RAW DATA

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID	obj_ID
2	1.24E+18	135.6891	32.49463	23.87882	22.2753	20.39501	19.16573	18.79371	3606	301	2	79	6.54E+18	GALAXY	0.634794	5812	56354	171
3	1.24E+18	144.8261	31.27418	24.77759	22.83188	22.58444	21.16812	21.61427	4518	301	5	119	1.18E+19	GALAXY	0.779136	10445	58158	427
4	1.24E+18	142.1888	35.58244	25.26307	22.66389	20.60976	19.34857	18.94827	3606	301	2	120	5.15E+18	GALAXY	0.644195	4576	55592	299
5	1.24E+18	338.741	-0.40283	22.13682	23.77656	21.61162	20.50454	19.2501	4192	301	3	214	1.03E+19	GALAXY	0.932346	9149	58039	775
6	1.24E+18	345.2826	21.18387	19.43718	17.58028	16.49747	15.97711	15.54461	8102	301	3	137	6.89E+18	GALAXY	0.116123	6121	56187	842
7	1.24E+18	340.9951	20.58948	23.48827	23.33776	21.32195	20.25615	19.54544	8102	301	3	110	5.66E+18	QSO	1.424659	5026	55855	741
8	1.24E+18	23.23493	11.41819	21.46973	21.17624	20.92829	20.60826	20.42573	7773	301	2	462	1.25E+19	QSO	0.586455	11069	58456	113
9	1.24E+18	5.433176	12.06519	22.24979	22.02172	20.34126	19.48794	18.84999	7773	301	2	346	6.96E+18	GALAXY	0.477009	6183	56210	15
10	1.24E+18	200.2905	47.1994	24.40286	22.35669	20.61032	19.4649	18.95852	3716	301	5	108	7.46E+18	GALAXY	0.660012	6625	56386	719
11	1.24E+18	39.14969	28.10284	21.74669	20.03493	19.17553	18.81823	18.65422	5934	301	4	122	2.75E+18	STAR	-7.9E-06	2444	54082	232
12	1.24E+18	328.0921	18.22031	25.77163	22.52042	20.63884	19.78071	19.05765	8102	301	3	27	5.65E+18	GALAXY	0.459596	5020	55852	525
13	1.24E+18	243.9866	25.73828	23.76761	23.79969	20.98318	19.80745	19.45579	3927	301	4	112	5.32E+18	GALAXY	0.591409	4727	55693	855
14	1.24E+18	345.8019	32.67287	23.17274	20.14496	19.41948	19.22034	18.89359	8157	301	2	38	7.32E+18	STAR	7.18E-05	6504	56540	574
15	1.24E+18	331.502	10.0358	20.8294	18.75091	17.51118	17.01631	16.62772	7773	301	2	123	5.7E+18	GALAXY	0.152194	5065	55739	200
16	1.24E+18	344.9848	-0.35262	23.20911	22.79291	22.08589	21.86282	21.8512	4192	301	3	255	1.04E+19	GALAXY	0.81816	9215	57682	796
17	1.24E+18	244.8245	25.15456	24.8868	22.13311	20.44728	19.49171	18.9747	3927	301	4	118	5.32E+18	GALAXY	0.484929	4727	55693	23
18	1.24E+18	353.2015	3.080796	24.5489	21.44267	20.95315	20.7936	20.48442	7712	301	5	284	4.82E+18	STAR	-0.00043	4283	55864	178
19	1.24E+18	1.494389	3.291746	20.38562	20.40514	20.29996	20.05918	19.89044	7712	301	5	339	9.84E+18	QSO	2.031528	8743	57663	295
20	1.24E+18	14.38314	3.214326	21.82154	20.5573	19.94918	19.76057	19.55514	7712	301	5	425	9.86E+18	STAR	-0.00044	8753	57373	258
21	1.24E+18	167.1317	67.33994	20.48292	18.67807	17.6168	17.11936	16.73351	1412	301	5	124	5.52E+17	GALAXY	0.111588	490	51929	613
22	1.24E+18	171.9754	67.74745	22.13367	20.84772	18.96537	18.31696	17.98124	1412	301	5	136	5.53E+17	GALAXY	0.374756	491	51942	572
23	1.24E+18	144.7853	46.8265	24.54793	22.33601	20.92259	19.87177	19.16934	2821	301	2	28	8.24E+18	STAR	-0.00012	7316	56710	756
24	1.24E+18	145.273	46.96013	25.44243	20.77028	19.6617	19.08481	18.83176	2821	301	2	30	9.39E+17	GALAXY	0.66231	834	52316	115
25	1.24E+18	145.883	47.30048	21.73992	21.53095	21.26763	21.36257	21.15861	2821	301	2	33	8.23E+18	QSO	2.07568	7308	56709	596
26	1.24E+18	241.4263	27.22469	18.88323	17.54229	17.01789	16.75376	16.72259	3927	301	4	94	1.57E+18	GALAXY	0.032081	1393	52824	368

Raw Data Examples

PREPROCESSING

Drop Identifier Columns

- Removed 9 non-physical columns (obj_ID, run_ID, rerun_ID, cam_col, field_ID, spec_obj_ID, plate, MJD, fiber_ID). Retained 8 features + class label.

Remove Missing Values

- Dropped any rows with NaN values. Dataset was clean — 100,000 rows preserved after this step.

Stratified 60/20/20 Split

- Train: ~60,000 rows | Validation: ~20,000 | Test: ~20,000. Stratified sampling preserves class ratios (~59.4% GALAXY / 21.6% STAR / 19.0% QSO) in every split. Random seed = 4700 for reproducibility.

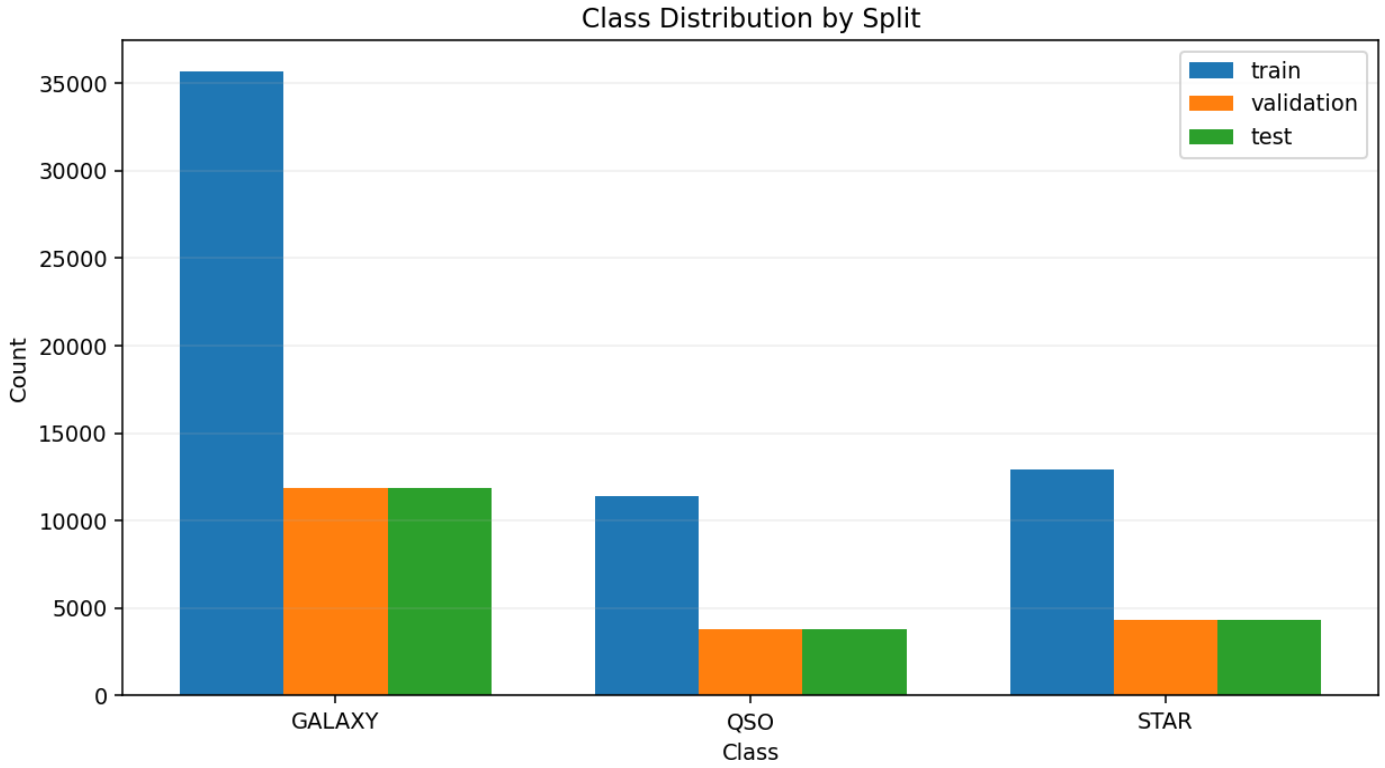
Z-Score Scaling

- Each feature scaled to mean=0, std=1. Scaler fit on training set only — same parameters then applied to validation and test sets. Prevents data leakage. Parameters saved to scaler_params.json for future use.

Output

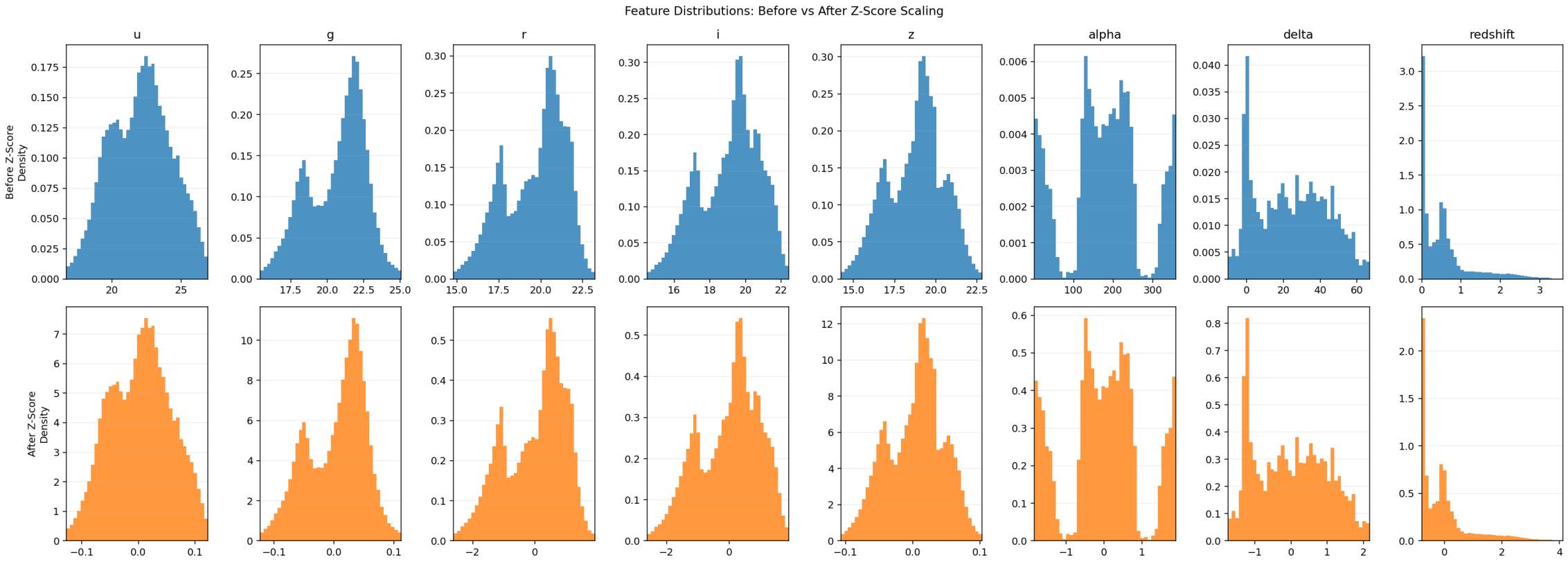
- train.csv, validation.csv, test.csv | raw_data.xlsx, preprocessed_data.xlsx | scaler_params.json

PREPROCESSING



The class proportions are consistent across training, validation, and test sets, confirming that stratified sampling was successfully applied.

PREPROCESSING



Before normalization, feature values vary significantly in scale.

After applying z-score normalization, all features are centered around zero with similar variance.

PREPROCESSING

	A	B	C	D	E	F	G	H	I	J
1	split	u	g	r	i	z	alpha	delta	redshift	class
2	train	0.069983	0.057892	1.587586	1.187808	0.073624	-0.33892	0.361512	0.278156	GALAXY
3	train	0.081832	0.05379	0.521554	0.15109	0.008497	-0.36629	0.580537	0.093136	GALAXY
4	train	0.005531	0.080958	1.062409	0.809723	0.015871	1.67348	-1.2489	0.488223	GALAXY
5	train	-0.06036	-0.07034	-1.69847	-1.76985	-0.07465	1.741366	-0.15146	-0.63091	GALAXY
6	train	0.038515	0.070244	0.90603	0.668198	0.023085	1.696872	-0.18168	1.163239	QSO
7	train	0.060837	0.046289	0.521856	0.217371	0.008748	0.236672	1.171126	0.114824	GALAXY
8	train	-0.00399	-0.0104	-0.25272	-0.15108	0.001314	-1.43561	0.200286	-0.79014	STAR
9	train	0.094244	0.050287	0.537253	0.397309	0.011169	1.562967	-0.30213	-0.15997	GALAXY
10	train	0.045333	0.081523	0.723145	0.412544	0.020895	0.69014	0.080075	0.020762	GALAXY
11	train	0.030814	-0.00771	-0.12102	0.078029	0.007162	1.746755	0.432619	-0.79003	STAR
12	train	-0.02638	-0.04175	-1.15122	-1.17775	-0.04819	1.598355	-0.71822	-0.58145	GALAXY
13	train	0.031702	0.05694	1.318444	1.583624	0.079412	1.738276	-1.24635	0.331662	GALAXY

Preprocessed Data Example

FEATURE EXTRACTION & DESCRIPTION

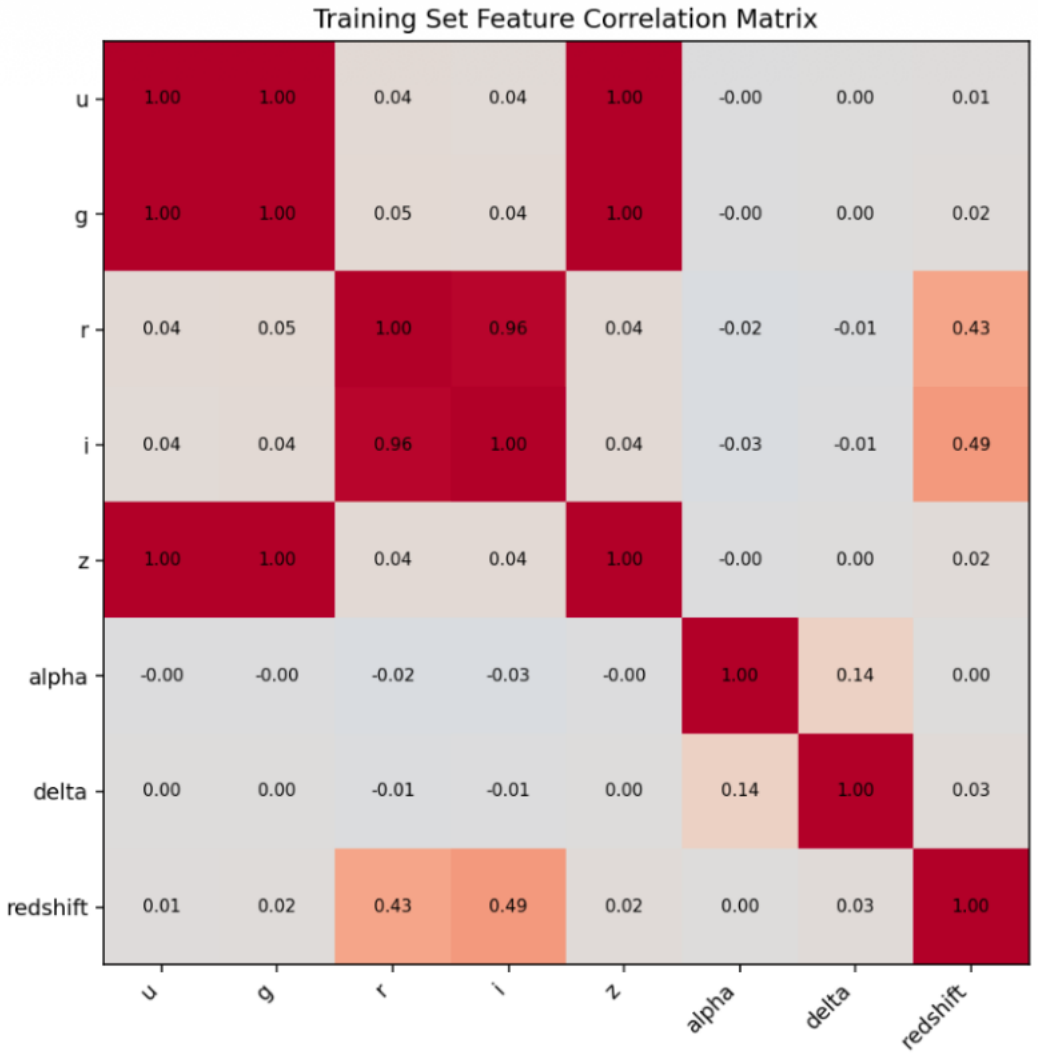
What was done:

1. Computed 8×8 Pearson correlation matrix on training features → revealed high collinearity among u/g/z filters and between u/g/z and redshift
2. Fit full PCA on training set, selected top 5 PCs to exceed 95% variance threshold (actual: 99.55%)
3. Applied PCA transform to all splits using training-fit parameters only

Parameters:

- Variance threshold: 95%
- Components selected: 5
- Cumulative variance (5 PCs): 99.55%

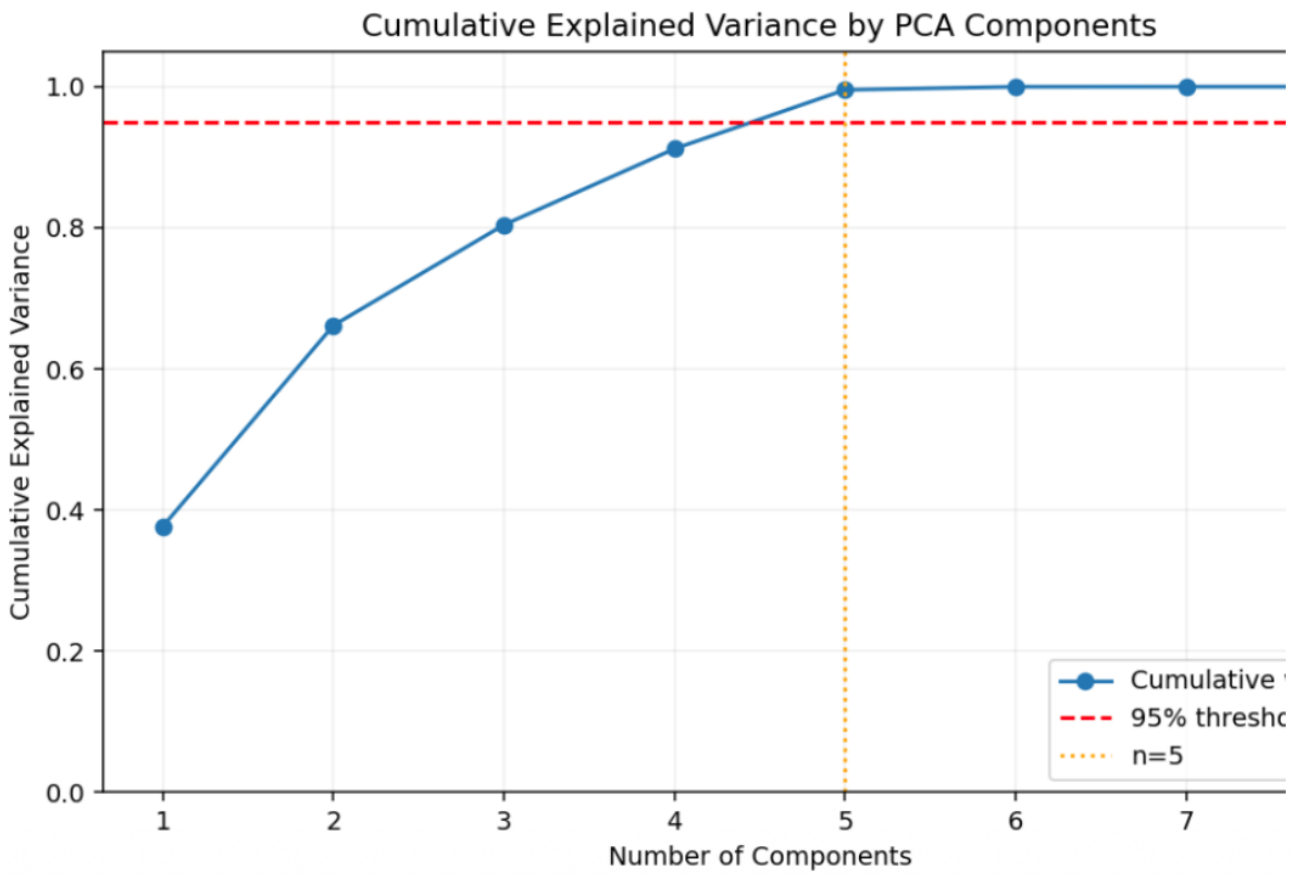
FEATURE EXTRACTION & DESCRIPTION



Correlation Heatmap

- Strong correlation between **u, g, z (~1.0)**
- Strong correlation between **r and I (~0.96)**
- Weak correlation with **alpha, delta**
- Indicates redundancy in original features

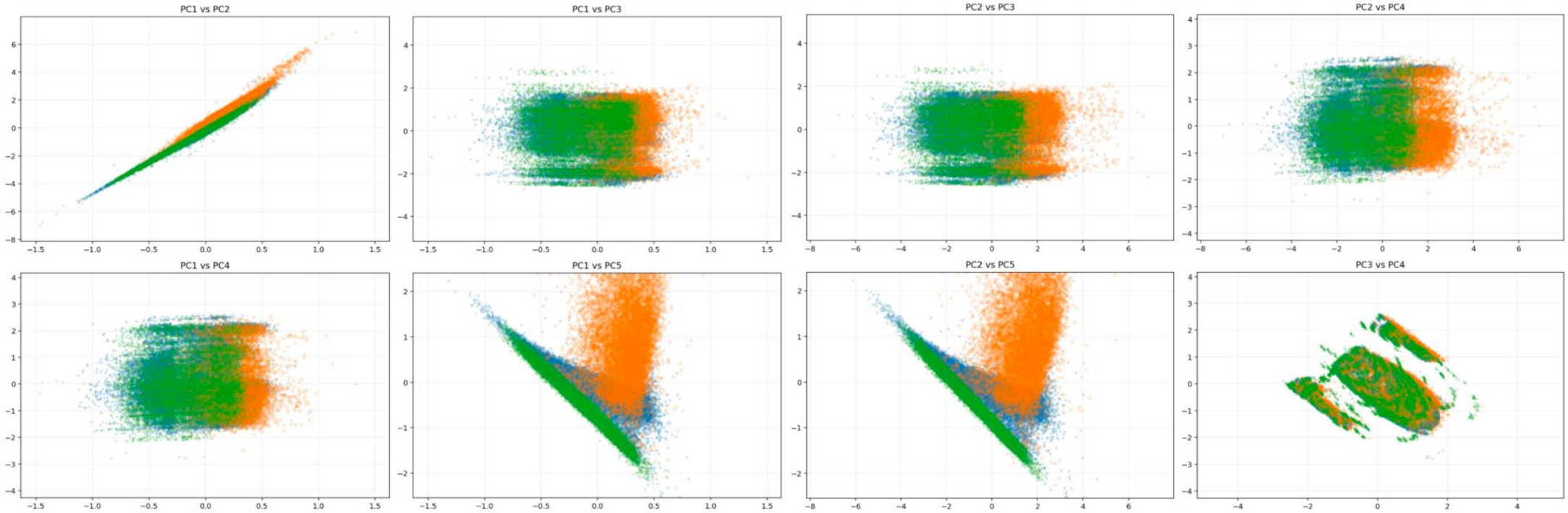
FEATURE EXTRACTION & DESCRIPTION



Cumulative Variance

- 5 components capture ≥95% variance
- Remaining components add very little information
- Selected optimal dimensionality = 5

FEATURE EXTRACTION & DESCRIPTION – PCA Scatter Plots



Data projected into lower-dimensional space. Shows some separation between: GALAXY, STAR, QSO.

FEATURE EXTRACTION & DESCRIPTION

1	split	PC1	PC2	PC3	PC4	PC5	class
2	train	0.40569	1.841957	-0.00222	-0.42754	-0.67769	GALAXY
3	train	0.15743	0.454934	0.14999	-0.65101	-0.19148	GALAXY
4	train	0.258896	1.334261	0.299814	2.100002	-0.0923	GALAXY
5	train	-0.50213	-2.46684	1.117944	1.273476	0.521743	GALAXY
6	train	0.29147	1.448632	1.126313	1.336542	0.506451	QSO
7	train	0.148642	0.495471	0.992249	-0.63514	-0.25397	GALAXY
8	train	-0.08784	-0.56965	-0.91907	-1.1447	-0.55959	STAR
9	train	0.162394	0.459183	0.857667	1.355558	-0.44611	GALAXY
10	train	0.193801	0.686064	0.5256	0.466198	-0.3749	GALAXY
11	train	-0.04106	-0.4224	1.474773	0.966153	-0.74823	STAR
12	train	-0.33863	-1.73951	0.60378	1.598024	0.258315	GALAXY
13	train	0.3869	1.901792	0.32198	2.17627	-0.5377	GALAXY
14	train	0.124594	0.330828	0.506805	0.485345	-0.3483	GALAXY
15	train	0.180664	0.639788	0.439135	2.121753	-1.16183	STAR
16	train	0.209992	1.503986	-1.89699	-0.60339	1.615201	QSO
17	train	-0.31681	-1.64892	1.462512	-1.65193	-0.0553	GALAXY
18	train	0.139858	0.352985	0.501291	-0.98855	-1.16889	STAR
19	train	0.070663	0.059336	0.594196	-1.05364	0.004179	GALAXY
20	train	0.036942	-0.17426	-0.92883	0.412532	-0.00357	GALAXY

Extracted Features (Excel Output)

- Enables efficient input for machine learning models
- Data is now lower-dimensional but retains key structure
- Generated dataset with:
  - PC1, PC2, PC3, PC4, PC5
  - Class labels preserved
  - Train/Validation/Test splits maintained

DESCRIPTION OF THE MODEL

Four classifiers are trained and compared:

1. Artificial Neural Network (ANN)
2. Support Vector Machine (SVM)
3. Decision Tree (DT)
4. K-Nearest Neighbors (K-NN)

All models are trained on PC1-PC5 extracted features from the training split. Performance is evaluated on the validation and test splits.

The pipeline follows strict no-leakage discipline. All preprocessing and feature transformations are fit exclusively on training data before being applied to validation and test sets.

DESCRIPTION OF THE MODEL

Model	Algorithm	Architecture / Key Parameters	Sklearn Class
ANN	Artificial Neural Network	Hidden layers: Dense (64, 32) neurons | Activation: ReLU | Optimizer: Adam | Max iterations: 500	MLPClassifier(hidden_layer_sizes=(64,32))
SVM	Support Vector Machine	Kernel: Linear | Multi-class: One-vs-Rest | Max iterations: 5,000 | Calibration: 3-fold CV	LinearSVC + CalibratedClassifierCV
DT	Decision Tree	Criterion: Gini impurity | Max depth: None (fully pure) | Min samples split: 2	DecisionTreeClassifier()
K-NN	K-Nearest Neighbors	K = 7 neighbors | Distance: Euclidean (L2) | Weights: Uniform | Input: 5 PCA components	KNeighborsClassifier(n_neighbors=7)

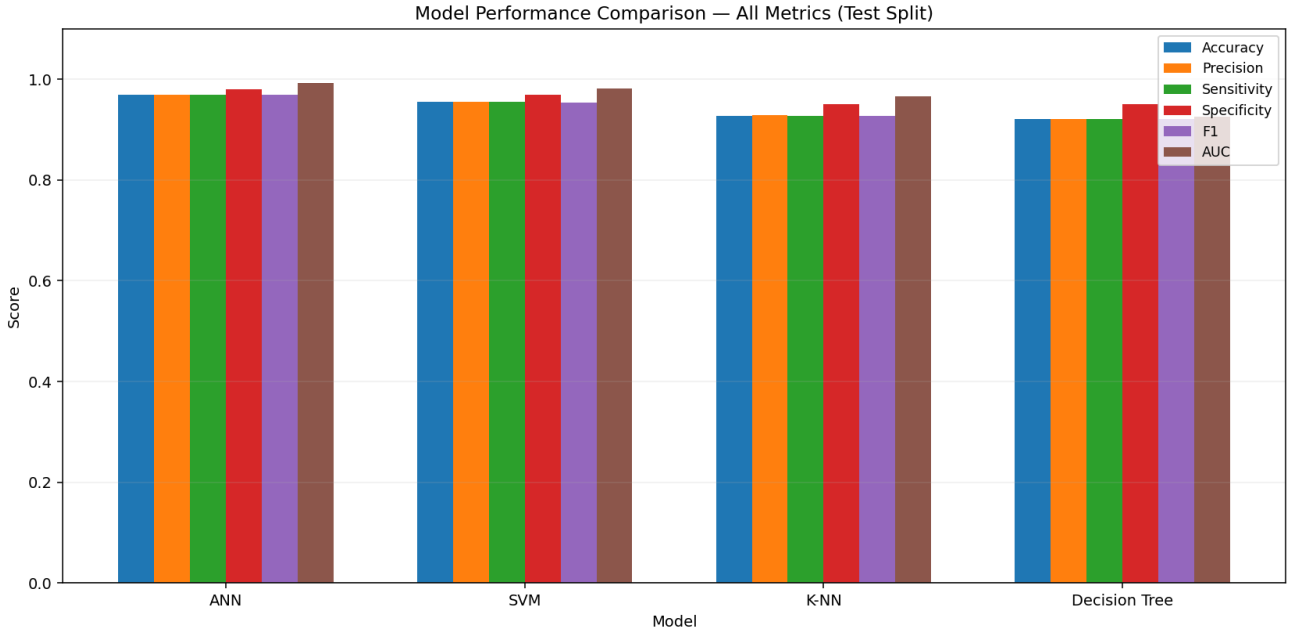
PERFORMANCE OF THE MODEL

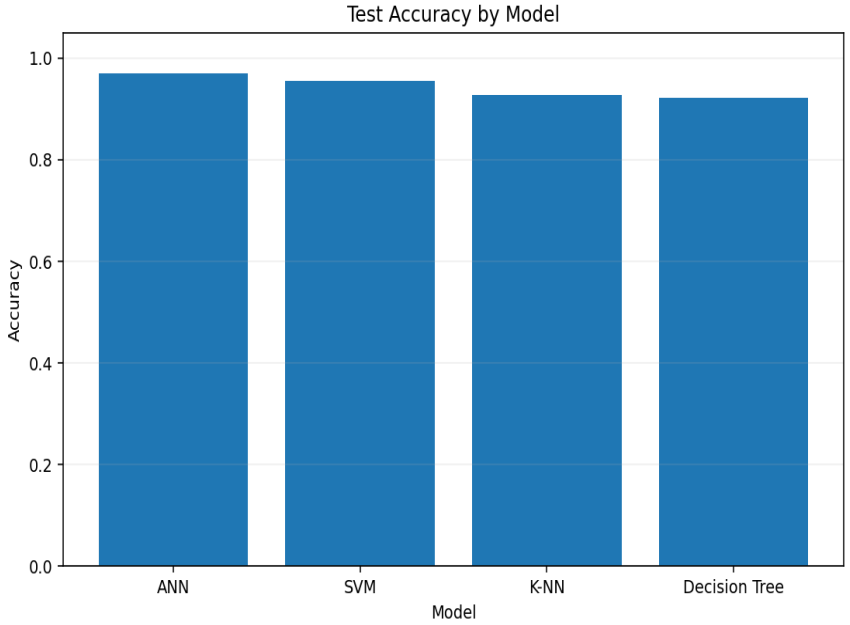
Model	Test Acc.	Test F1	AUC	Sensitivity	Specificity	Notes
ANN (64→32)	97.0%	0.970	0.998	97.0%	98.5%	Best overall. lowest train/test gap
SVM (Linear)	95.5%	0.954	0.993	95.5%	97.8%	Strong linear separator in PC space
K-NN (k=7)	92.8%	0.927	0.983	92.8%	96.4%	No explicit training; distance-based
DT (unlimited)	92.2%	0.922	0.922	92.2%	96.1%	100% train acc. → overfitting

Key Findings

- ANN achieves best test accuracy (97.0%) and highest AUC (0.998) — generalizes well with minimal overfitting
- Decision Tree overfits severely: 100% training accuracy vs. 92.2% test accuracy — classic depth-unlimited DT behavior
- Redshift is the dominant feature — STARs cluster near redshift ≈ 0 while GALAXYs and QSOs have progressively higher values

PERFORMANCE OF THE MODEL

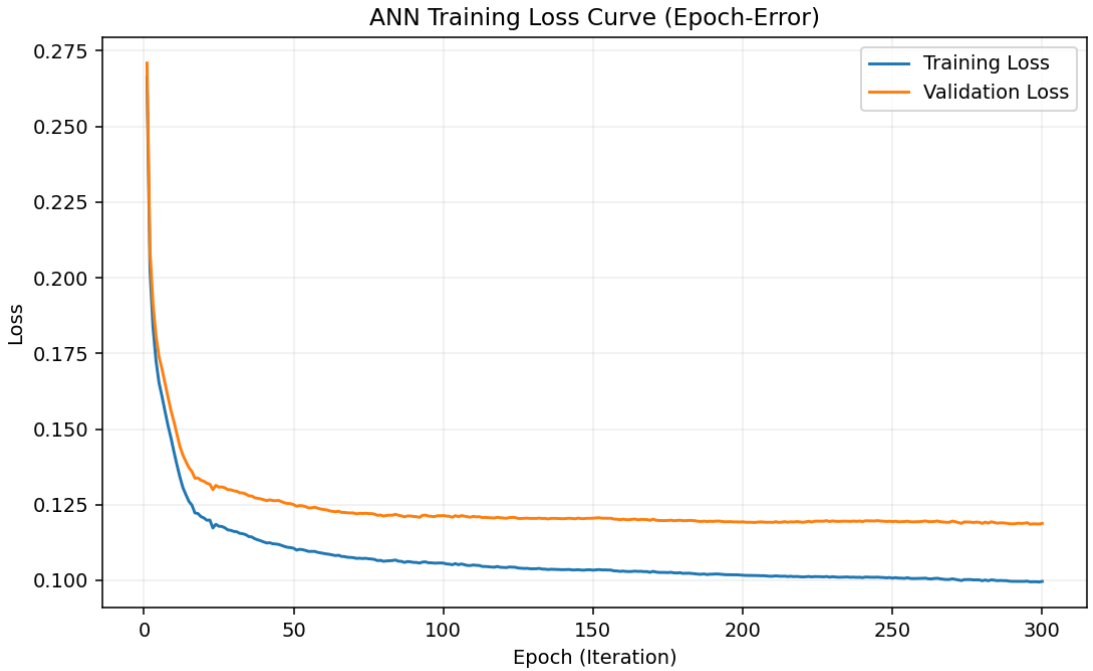




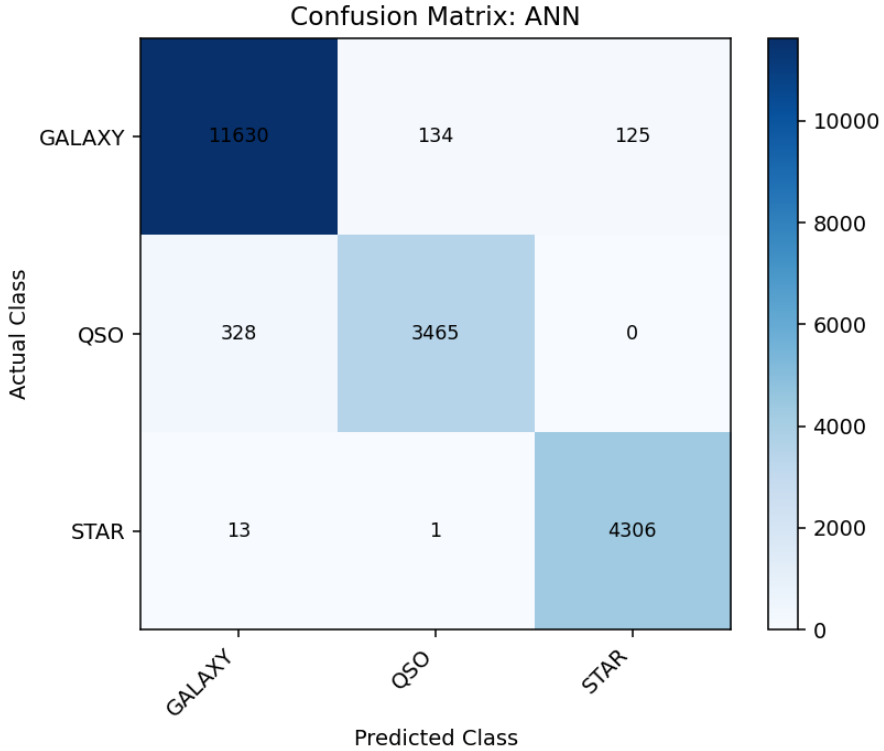
ANN leads across all 6 metrics

PERFORMANCE OF THE MODEL – ANN

- ANN train accuracy ~96.9% vs test 97.0% → **no overfitting**, excellent generalization

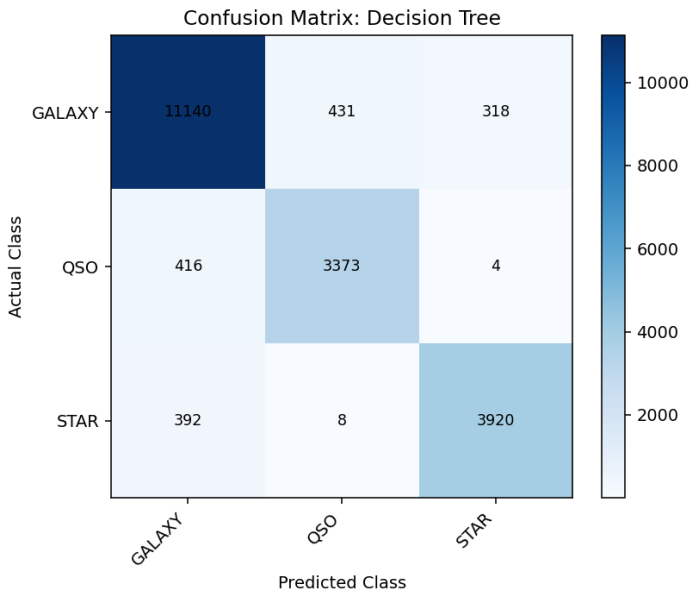


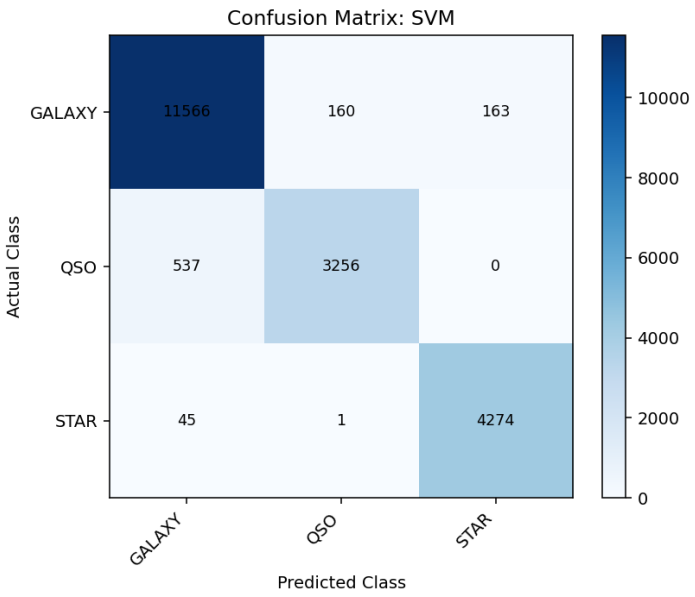
ANN training loss converges smoothly — no instability

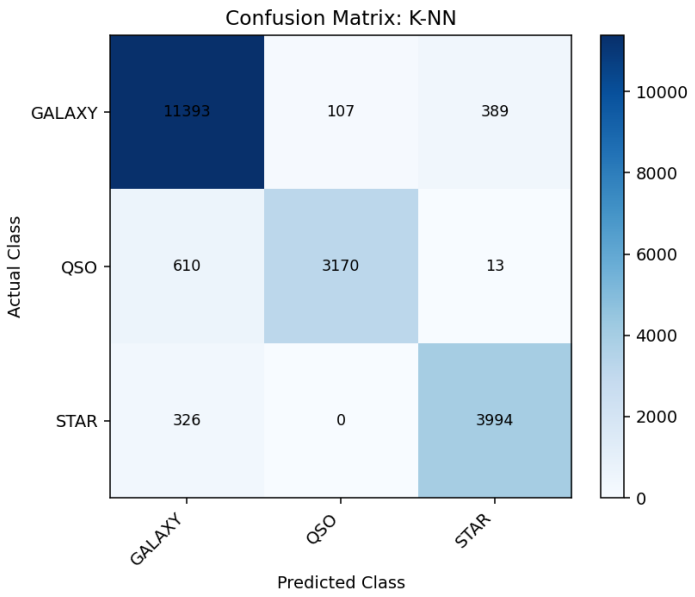


ANN confusion matrix (test set) — very few misclassifications

PERFORMANCE OF THE MODEL – Other Models







- Decision Tree: train accuracy **100%** vs test **92.2%** → clear **overfitting**
- All models struggle most with GALAXY/QSO boundary (similar photometric profiles)
- **Best model: ANN** — highest scores across all metrics on test set